

Aether Dark

Dual-voice drone instrument · User manual

Aether Dark is a browser instrument: two analog-style voices, a shared Drift LFO, noise, tape print, echo, bloom, a patchbay, and Evolve — a slow climate layer so the sound can change over minutes without becoming a different patch.

Play it as a webpage (GitHub Pages <https> link) or as a downloaded index.html in desktop Chrome/Edge. Tap POWER once. MIDI (including the Akai MIDI Mix map) works in those desktop browsers. iPad: open the <https> link in Safari and keep the tab in front — do not open the file from the Files app preview. Web MIDI is not available on iOS. Support: ko-fi.com/tribalvoice

Not affiliated with Dubreq or Stylophone. Original Aether Series instrument.

1. Signal flow

Voice I and Voice II each mix a filtered saw/square with a triangle and a triangle sub, through a soft shaper and a state-variable-style biquad (LP/HP). Both voices hit the mix bus → echo → bloom → optional tape print → main out. Noise can join the mix or its own copy of Filter I / Filter II. Drift is a single-rate LFO with two waveforms. Evolve does not add an oscillator; it walks offsets around the panel values.

2. Drift LFO

One rate, two shapes. Destination toggles choose whether each voice's osc and filter listen to Wave I or Wave II.

Control	What it does
Wave I / Wave II	Stepped shapes including Off, triangle, square, saw, multiples. Click the knob to cycle.
Rate	0.02–70 Hz. The swell / pulse of the instrument.
OSC / FILTER dest	Up = Wave II, down = Wave I.
RATE jack	CV in, scales LFO rate.
RESET jack	Rising edge restarts both phases.
WAVE I / II jacks	CV out of the current LFO sample.

3. Evolve — climate, not a second LFO

Drift is the wave you already hear. Evolve is weather over a minute or several minutes so the next swell is not a photocopy of the last. Depth 0 leaves every factory patch unchanged.

3.1 Controls

Control	Range	Role
Time	45 s – 8 min (log)	How long the climate takes to lean to a new target, then pick another.
Depth	0–1, default 0	How far Evolve may stray from the panel knobs. Knobs do not move; Evolve is an offset on top.
Target	Tone / Weather / Motion / All	What it is allowed to touch.
Irregular	0–1, default 0	On each wrap of Wave I, dice a new offset for filter LFO depths and noise LFO depth. Separate from Time.
Loop / Once	toggle	Loop (down) keeps wandering. Once (up) takes one journey and holds.
CV jack	–1...+1	The current wander. Patch to cutoff or pitch if you want the climate as a cable.

3.2 Targets

Target	Moves
Tone	Both cutoffs and both X-Wave positions (timbre tilt). Never pitch or volume.
Weather	Noise level, and a tilt of the noise colour filter / tape wear feel.
Motion	Drift Rate and Noise LFO Rate $\times$ (surf speeds up or dies).
All	Tone + Weather. Rate is left alone so the pulse stays your pulse.

3.3 How it moves

Not a triangle. A slow random walk: Evolve eases toward a hidden aim, then chooses a new aim. Time is the scale of that journey. Depth scales how far from your cutoff / X-Wave / noise it may go. Turning a knob still means that knob — the climate breathes around it.

Irregular is the “each wave is a different wave” control. 0 = identical cycles. Noon = set-to-set change of swell height. Full = almost sample-and-hold on how far this crash opens.

3.4 Listening examples

Shoreline + Time $\sim$ 2 min, Depth $\sim$ 30%, Target Weather — shore stays shore, hiss fattens, print ages.

Breakers + Irregular at noon — crashes not the same height.

Chapel Fifth + Target Tone, Depth low, Once — the fifth slowly blooms and stays open.

Factory patch Long Climate is Tide/shore language with All + Irregular already on. Deep Tide also has a gentle All wander. Leave them running; do not judge in ten seconds.

3.5 What Evolve does not do

No second waveform bank. No tempo sync. No morphing one factory patch into another. No Main Out automation.

4. Voices

Each voice: Osc Mix (X-Wave $\leftrightarrow$ sub), Sub $-1/-2$, X-Wave, LFO $\rightarrow$ pitch or X-Wave, Pitch + tuner, Filter resonance / LP-HP / LFO depth, Cutoff, Volume. Voice II adds Hard Sync, Pitch Link, and Offset in semitones. Jacks: PITCH, PWM, CUTOFF, OUT.

Dark / Dusk sits on the chassis rail and is saved with the patch. Dark is the three-way clone-shaped blend (square–triangle–saw). Dusk inserts sine in the middle of the travel (square–triangle–sine–saw). The sine oscillator is already in the voice; Dark simply turns its gain to zero. Factory patches load as Dark unless you saved them otherwise.

5. Noise

Control	What you hear
Level	How much noise. Independent of voice volume when Dest is a filter.
Colour	White / Pink / Brown = textures. S&H = stepped chunks of hiss (a new loudness each clock), not a DC voltage.
Dest	Off / Mix / Filter I / Filter II. Filter dest uses a copy of that voice’s cutoff, Q and LP/HP.
LFO Depth	Swell of noise level from Wave I. 0 = flat.
LFO Rate $\times$	$0.25\times-4\times$ Drift Rate.
Phase Offset	$0-360^\circ$. 180° puts the crash in the trough of the voice swell.
OUT	Audio, and S&H CV when Colour is S&H.

S&H clock is Drift Rate $\times$ Noise LFO Rate. Only Embers in the original factory set used it; after the DC fix it is audible as irregular grit.

6. Mixer, echo, bloom, tape

Delay D/W, Time, Repeats, echo LFO depth and Wave I/II source. Bloom algorithm. Dry voices sit in the centre; Bloom's wet tail is stereo. Tape: Print (level-matched sat — Print off only mutes the wet path, it does not unload the tape graph), Sat, Wow, Wear.

REC captures the printed mix. SAVE writes a 16-bit mono WAV. PLAY auditions the last take; with POWER off it plays the take only and does not start the drones. For a production stereo bounce, record the browser output in a DAW (Audacity on Windows: WASAPI loopback), not SAVE.

7. Patchbay

Click an output, click an input. Click a cable to remove it. Clear Cables in the toolbar. Audio: voice OUT and noise OUT. CV: LFO waves, Evolve CV, RATE and RESET, voice PITCH / PWM / CUTOFF.

8. MIDI

Generic map: CC1 rate, CC7/11 level, 71/74 Voice I res/cut, 76/77 Voice II, 91 bloom, 94/72/73 echo. Notes tune the drones. Program change loads factory patches in list order.

Map: Akai MIDI Mix uses the Mix factory CCs. Mute/Rec notes are buttons, not pitches. Do not hit Send All unless you want every parameter slammed. Bank left/right step factory patches if they send notes 25/26.

9. Factory patches (selected)

Patch	Idea
Ambient Wash / Stereo Hall / Chapel Fifth	Slow linked drones, little or no noise.
North Wind / Gale / Shoreline / Breakers	Noise as weather. Breakers pulses brown through Filter I, 180° offset.
Rain / Thunder Bank / Night Insects / Embers	Detail weather. Embers = S&H into Filter II.
Deep Tide / Long Climate / Fog / Glacier	Leave running. Climate and Tide wander under Evolve.

10. Hosting and files

One GitHub repository holds everything. Name the instrument index.html so Pages can serve it. Enable Settings → Pages → deploy from the main branch, folder / (root). The play URL is <https://YOURUSER.github.io/REPO/>.

iPad and any browser: that https link. PC offline: download index.html from the repo (or a Release) and open it in Chrome. Same file, two ways to reach it. Put this PDF in the repo root as Aether-Dark-User-Manual.pdf so the Manual link on the page resolves.

Do not use the iPad Files preview. If knobs are dead, close that preview and open the Pages URL in Safari. The orange banner can be dismissed; it also hides after POWER succeeds, and on http/https.

11. Support and patches

The instrument is free. Ko-fi: ko-fi.com/tribalvoice (toolbar chip and footer). Randomise includes noise, Evolve and sometimes Dusk. Patches live in this browser's localStorage; Export JSON to move them between machines.